CLINICAL AUDIT REPORT

NeuroPulse XAI — Stroke CDSS (Research Prototype)

1. PATIENT PROFILE

Age / Sex **67 years / Male**	Systolic BP **165 mmHg**	Avg glucose **210.0 mg/dL**	BMI **31.4**
Total cholesterol **240.0 mg/dL**	Smoking **smokes**	Physical activity **2.0 / 10**	Onset to ER **2.2 h**

2. TRI-PILLAR PREDICTIONS

Stroke risk (10-year) **93.3%** Critical / Very High Risk	NIHSS severity **32.8 / 42** Severe (NIHSS 21-42)	Favorable outcome P(mRS 0-2) **14.8%** Lower likelihood of functional independence (mRS 3-6 possible)

3. TOP SHAP ATTRIBUTIONS (STROKE RISK MODEL)

FEATURE	SHAP (Φ)	DIRECTION
Average glucose (mg/dL)	+0.8307	Risk-increasing
Heart disease / AFib: yes	+0.7921	Risk-increasing
Age	+0.6740	Risk-increasing
Smoking: current smoker	+0.6391	Risk-increasing
Systolic BP (mmHg)	+0.5262	Risk-increasing
Physical activity (0-10)	+0.3841	Risk-increasing

4. COUNTERFACTUAL INTERVENTIONS

Blood pressure normalisation — ↓ 22.6 pp absolute risk reduction

165.0 mmHg → 120 mmHg (normotensive)

AHA guideline: first-line antihypertensive (ACEi/ARB or CCB). SBP <130 mmHg reduces cerebrovascular shear stress.

Glycaemic normalisation — ↓ 14.3 pp absolute risk reduction

210.0 mg/dL → 95 mg/dL (euglycaemic)

ADA/AHA target: HbA1c <7.0%, fasting glucose <100 mg/dL. Reduces microvascular and macrovascular angiopathy.

Tobacco cessation — ↓ 14.3 pp absolute risk reduction

smokes → non-smoker

Cessation reduces platelet hyperaggregability. Stroke hazard reduction observed over 2-5 years post-cessation.

Physical conditioning and weight reduction — ↓ 8.7 pp absolute risk reduction

Activity 2.0/10, BMI 31.4 → Activity 7.5/10 (≥150 min/wk), BMI 23.5

Aerobic exercise improves cerebral blood-flow autoregulation and lowers arterial stiffness.

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